

8 am

9 am

10 am

11 am

12 noon

1 pm

2 pm

3 pm

4 pm

5 pm

6 pm

FULL STACK

WEB DEVELOPMENT

→ Web - It refers to all the public websites or pages that users can access on their local computers and other devices through the internet. In web large amount of documents, data resources are interconnected and can be accessed via internet.

→ Web Development - It is a process/entity in which we create/develop websites and web applications.

Websites

- Static in nature
- Read Only mode.
- User is unable to manipulate information.
- Minimal user interaction available.
- e.g. Wikipedia

Web Apps

- Dynamic in nature
- Read and Write Mode
- User able to manipulate information.
- Users can interact in web Apps.
- e.g. LinkedIn, Facebook, ~~Google~~ Google

17

WEDNESDAY
JANUARY

WK 03 (017-348)

JANUARY 2018

M	T	W	T	F	S	S	M	T	W	T	F	S	S
1	2	3	4	5	6	7	8	9	10	11	12	13	14
15	16	17	18	19	20	21	22	23	24	25	26	27	28
29	30	31											

8 am → Web App

Web App

9 am

10 am Front End/UI

Back-End

Data Base Layer

11 am Front End Engineers

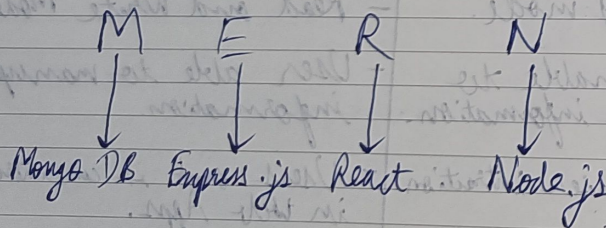
Back End Engineers

Data Base Engineers

12 noon User Interface /
Looks of Web
pagesBrain of Web App
/ Logical used in
Web AppManipulation / Interaction
of data, all data is
stored in data base2 pm HTML/CSS/JS
+ ReactPHP/Python/Javascript/
Django/Mongo DBNoSQL/SQL/
MongoDB

3 pm

4 pm ⇒ Hence A Full Stack Developer can
work on all the layers to develop
a Web Application (i.e. Front End, Back End,
and Data base).



2018 FEBRUARY

M	T	W	T	F	S	S	M	T	W	T	F	S	S
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WK 03 (018-347)

18

THURSDAY
JANUARY

→ Network :-

8 am

9 am

10 am

11 am

12 noon

1 pm

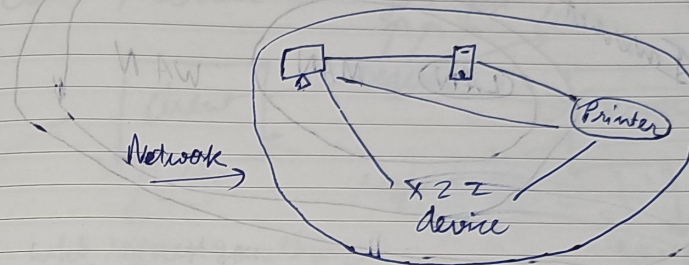
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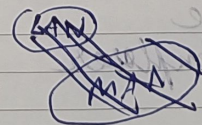
Network is a system in which all the devices
are interconnected (e.g. Computer, mobile, wifi,
printer) by which they can communicate,
share resources, etc.

→ Internet :- It is a global system or it is
network of networks consists of billion of
devices interconnected with each other.

• LAN :- Local Area Network (consist of rooms,
building)

• MAN :- Metropolitan Area Network (consist of ~~states~~ states,
cities)

• WAN :- Wide Area Network (consist of countries,
continents)



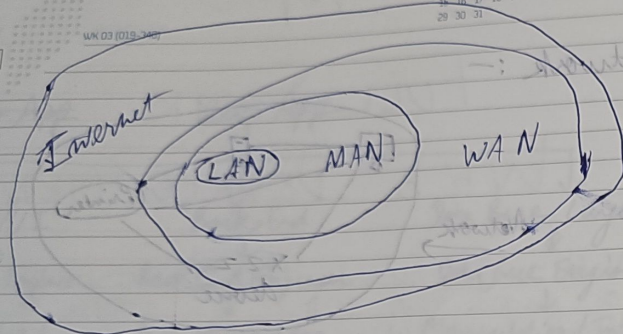
2018

19

FRIDAY
JANUARY

JANUARY 2018

M	T	W	T	F	S	S	M	T	W	T	F	S	S
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→ WWW :- ~~World Wide Web~~ World Wide Web. Interconnected documents, devices, resources can be accessed via Internet. It is a very small subset of Internet.

→ IP Address :- Address for identifying each device uniquely.

→ Browser :- It is a tool to access & access resources, websites, web apps. e.g. Chrome, Safari, Opera, etc.

→ Browser Engine :- It is a underline brain which makes it possible to render, execution.

e.g. Webkit → Safari
Gecko → Mozilla Firefox
Blink → Chrome
EdgeHTML → MS Edge
Trident → Internet Explorer

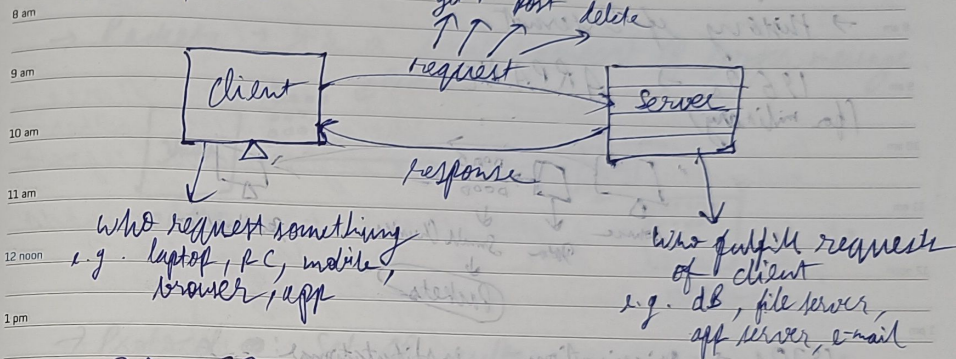
2018 FEBRUARY

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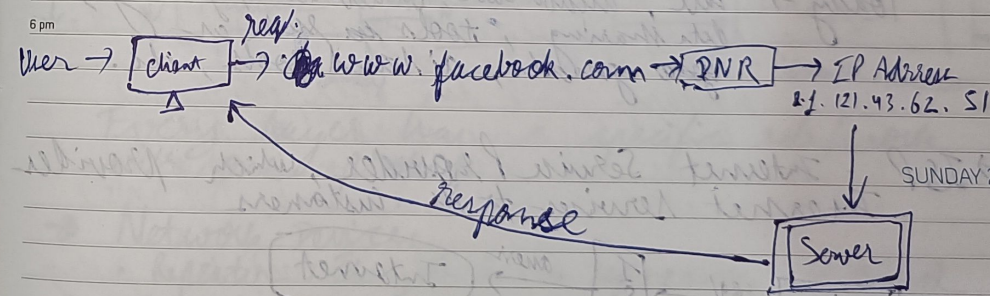
SATURDAY
JANUARY

→ Client / Server :-



→ DNS (Domain Name System) :- It convert domain name into IP Address. or DNR (Domain Name Resolution)

→ Domain Name - is an easy to remember name that's associated with a physical IP address on the Internet. Its always unique.



SUNDAY 21

2018

22

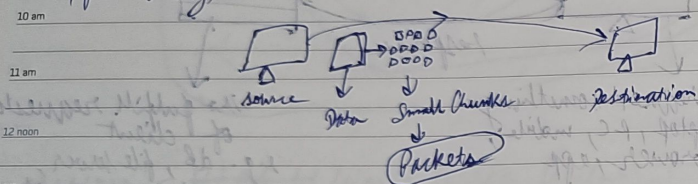
MONDAY
JANUARY

WK 04 (022-343)

JANUARY 2018

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8 am → History of Internet

9 am 1960s → ARPANET
(for military)

1970s → Organisations, institutional, ...

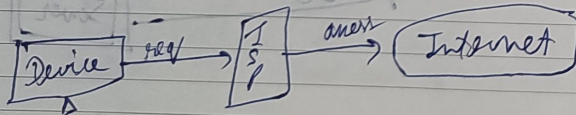
1980s → DNS, //www

1990s → Email, msg, Shopping

2000s → Facebook, Google, Amazon, mobile phone

Today → call, audio call, video call, gaming, data sharing, tools like such as ~~gms~~ gmail, ai based tools, etc.

→ ISP: Internet Service Provider, which provides Internet services to its customers



2018

2018 FEBRUARY

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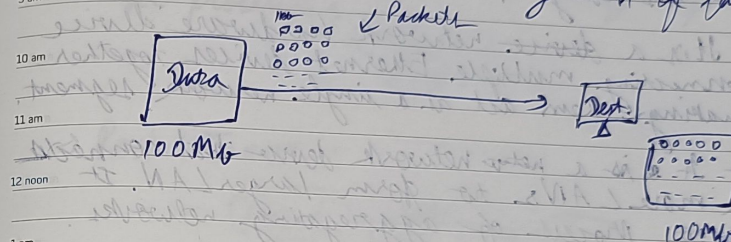
WK 04 (023-342)

23

TUESDAY
JANUARY

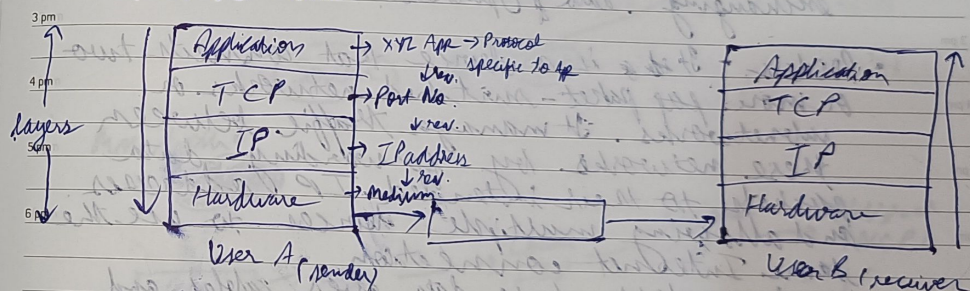
→ Protocol Stack

→ Packets → It is a small segment of larger message.



→ Protocol: Set of rules

→ Protocol Stack



Every layer has a specific role/work with some protocols

→ Network Devices

Repeater: Its job is to regenerate the signal over the same network before the signal becomes too weak or corrupted to extend the length/range to which the signal can be transmitted over the same network.

2018

24

WEDNESDAY
JANUARY

WK 04 (024-341)

JANUARY 2018						
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8 am

Its job is to amplify, regenerate the signals to increase its range.

9 am

10 am

• Hub - It is a device network hardware device for connecting multiple Ethernet devices together and making them act as a single network segment.

11 am

12 noon

• Bridge - It is a network device that connects two or more LANs to form larger LAN. It refers to process of aggregating networks.

1 pm

2 pm

• Switch - It connects devices in a network to each other, enabling them to talk by exchanging data & packets.

3 pm

4 pm

5 pm

6 pm

• Routers - It is a device that connects two or more packet-switched networks or subnetworks. It manages traffic between these networks by forwarding data packets to their intended IP addresses and allowing multiple devices to use the same Internet connection.

Wired routers share data over cables and create wired local area networks (LANs).

Wireless routers use antennas to share data and create wireless local area networks (WLANs).

• Gateway: (also called protocol converter) is a passage to connect two networks that may work upon different networking models. It is very complex.

2018 FEBRUARY

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WK 04 (025-340)

25

THURSDAY
JANUARY

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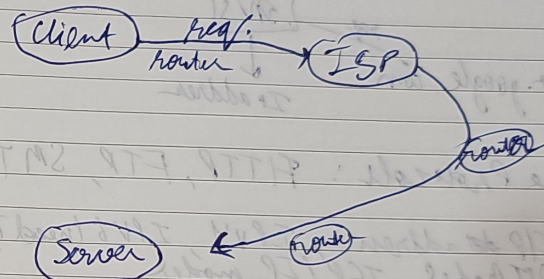
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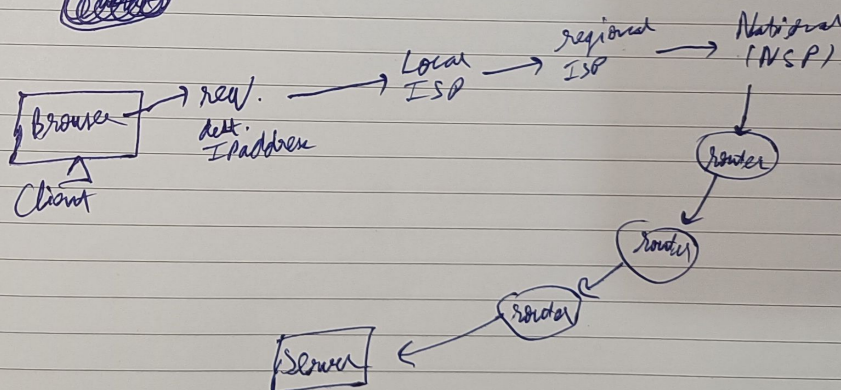
Modem - It is a network device that both modulates and demodulates analog carrier signals (called sine waves) for encoding and decoding digital information for processing.

modulate
demodulate

→ Packet Travel:



More depth:



2018

26

FRIDAY
JANUARY

WK 04 (026-339)

JANUARY 2018

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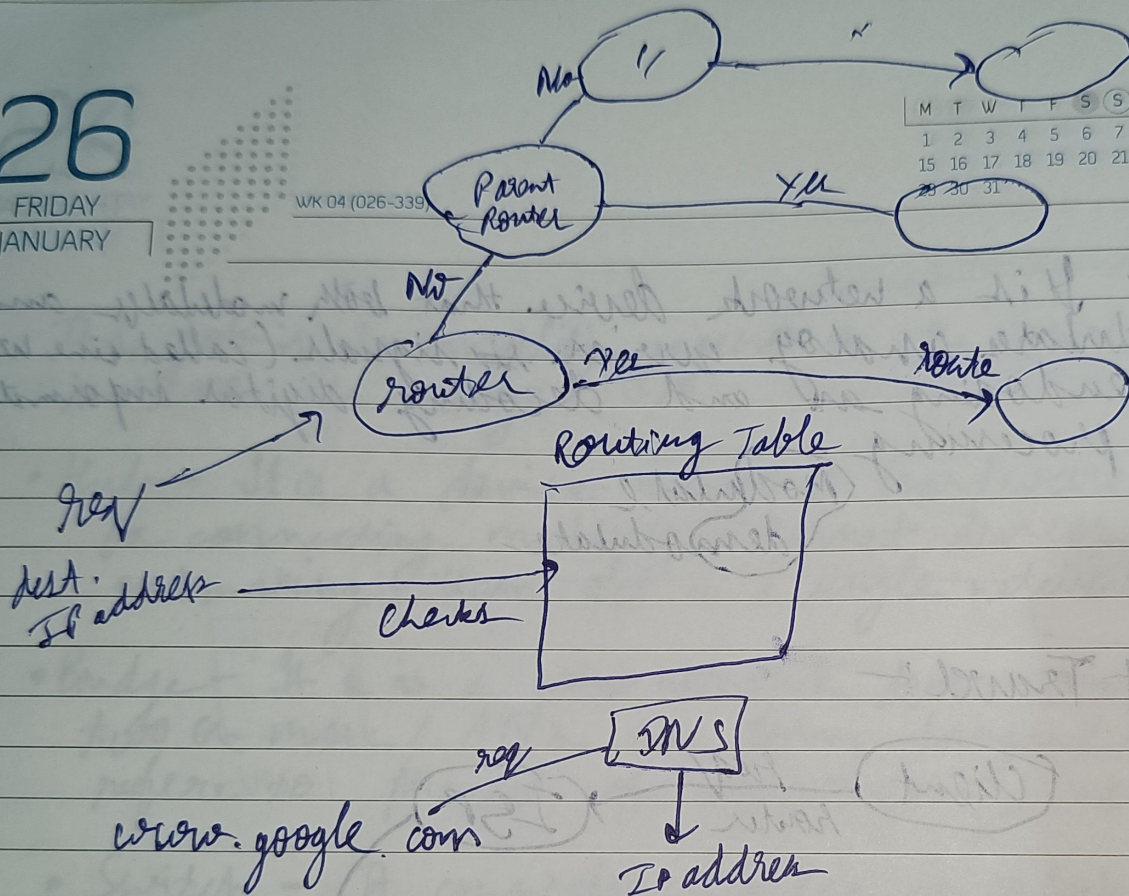
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~~Some~~ Some Protocols: HTTP, FTP, SMTP, etc.

H/w: IP address, IPV4, IPV6 (need?)
OSI Model, TCP/IP model

2018 FEBRUARY

M	T	W	T	F	S	S	M	T
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